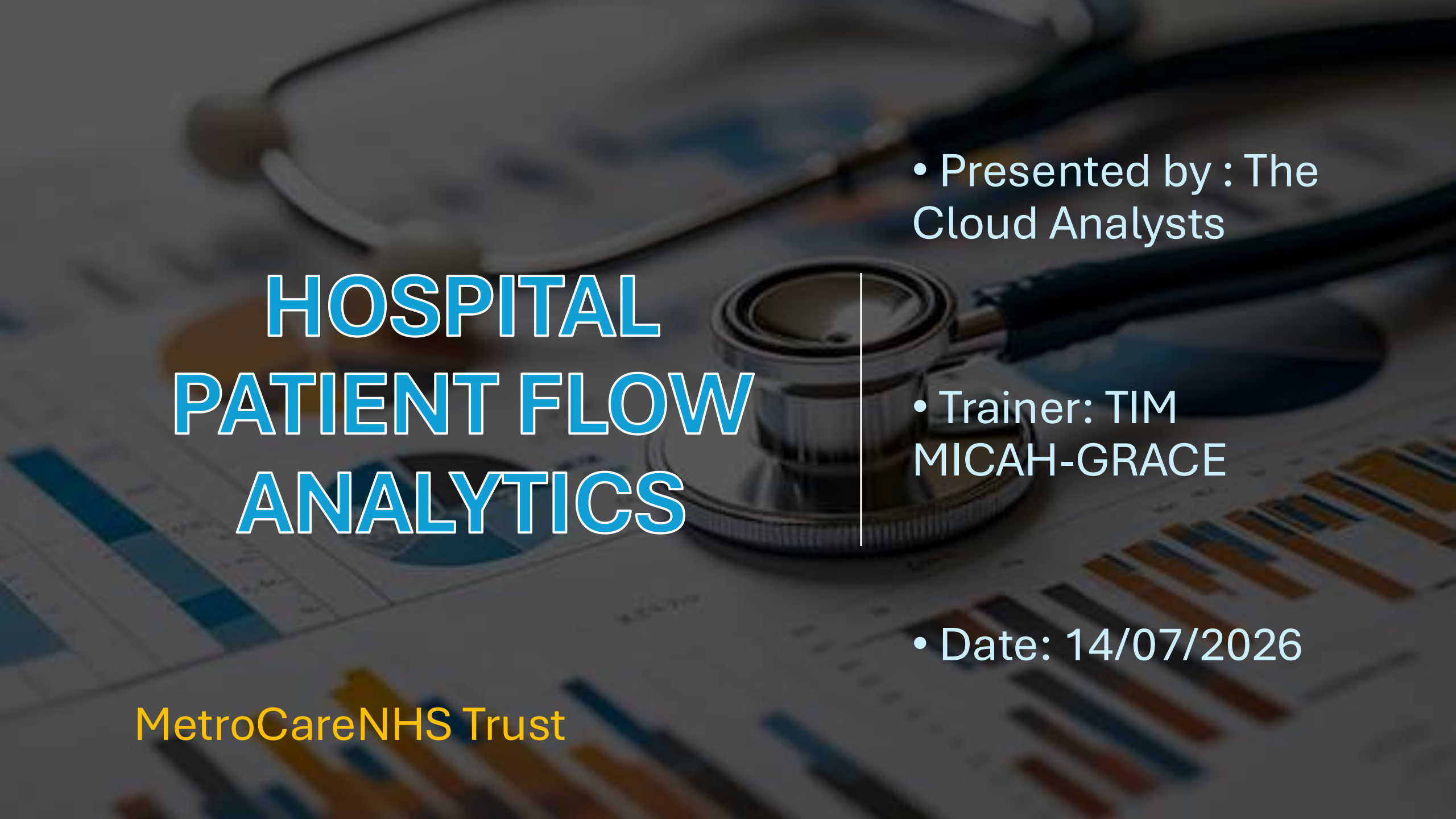




HOSPITAL PATIENT FLOW ANALYTICS

- Presented by : The Cloud Analysts

- Trainer: TIM
MICAH-GRACE

- Date: 14/07/2026

MetroCareNHS Trust

Introduction

MetroCare NHS is experiencing pressure on:

- Patient flow
- Patients waiting time
- Bed Availability



Objectives:

- Analyse waiting times by department
- Understand admission outcomes
- Investigate whether staffing levels affect delays
- Identify key causes of poor patient flow.
- Conclusion
- Recommendations



Dataset Overview and Methodology

Datasets used:

- **Admissions dataset**
 - Admission type
 - Waiting time
 - Length of stay
 - Outcome
 - Bed pressure score
- **Staffing dataset**
 - Date
 - Department
 - Staff on duty

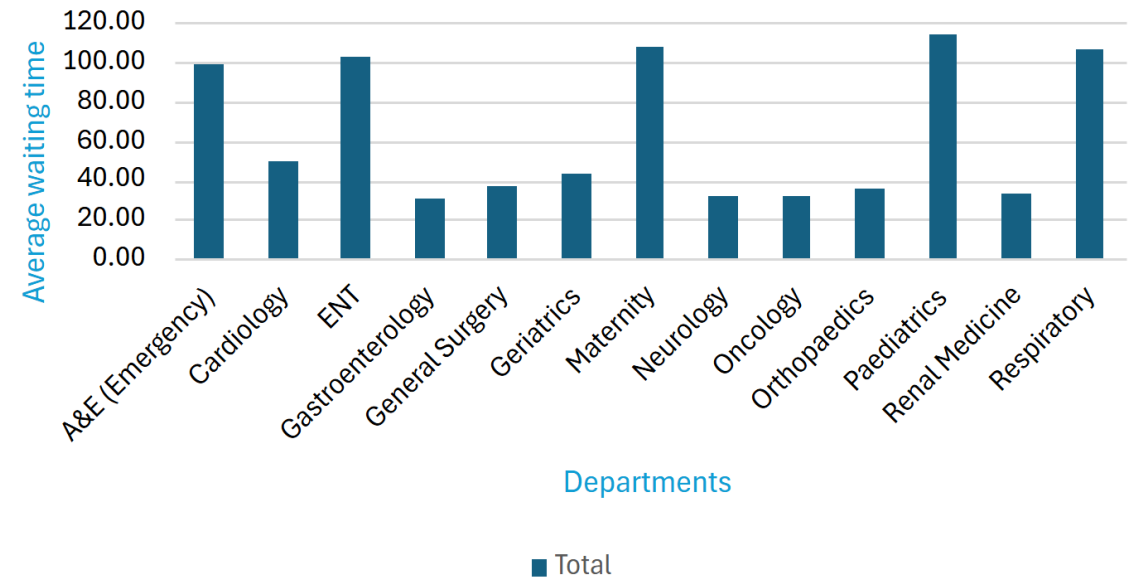
Methods:

- Data cleaning
- VLOOKUP/XLOOKUP to join staffing data
- PivotTables
- Data visualisations

Average Waiting time by department

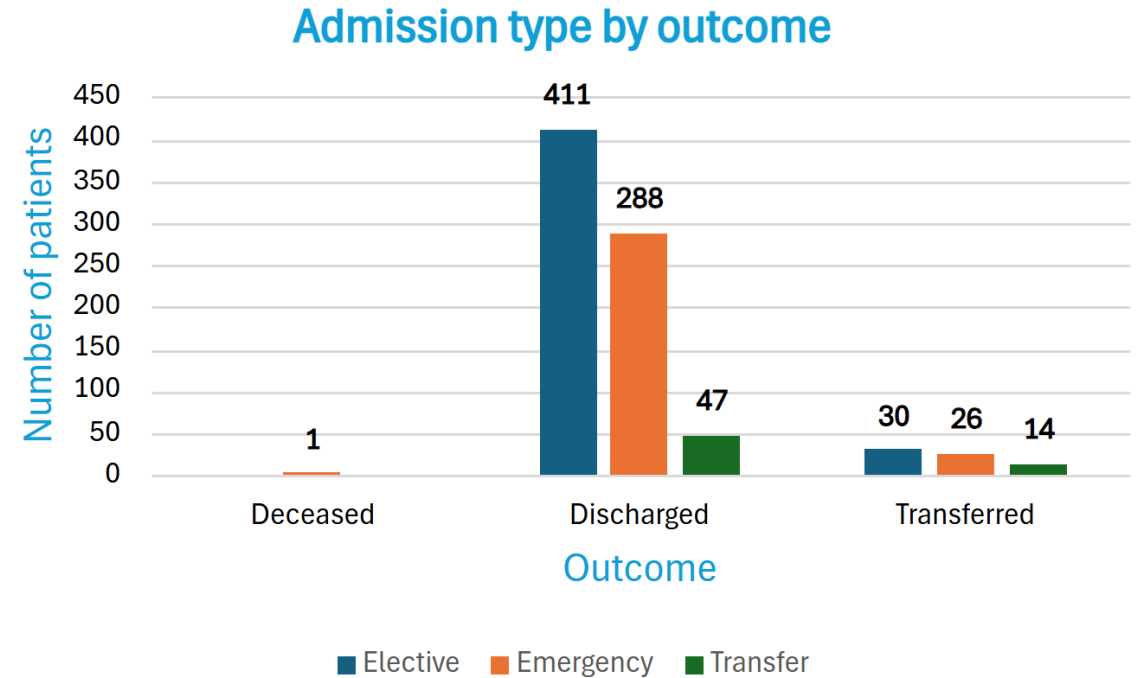
- All departments experience waiting times **above 20 minutes**, showing common pressure on patient flow.
- Paediatrics has the highest average waiting time (**114 minutes**).
- Maternity, Respiratory, ENT, and A&E also show significant delays.
- These departments should be prioritised for operational improvement.

Average waiting time by Department



Admission type by Outcome

- Total number admitted: **817 patients**
 - 746 patients discharged.
 - 70 patients transferred.
- Emergency admission recorded 1 deceased outcome.

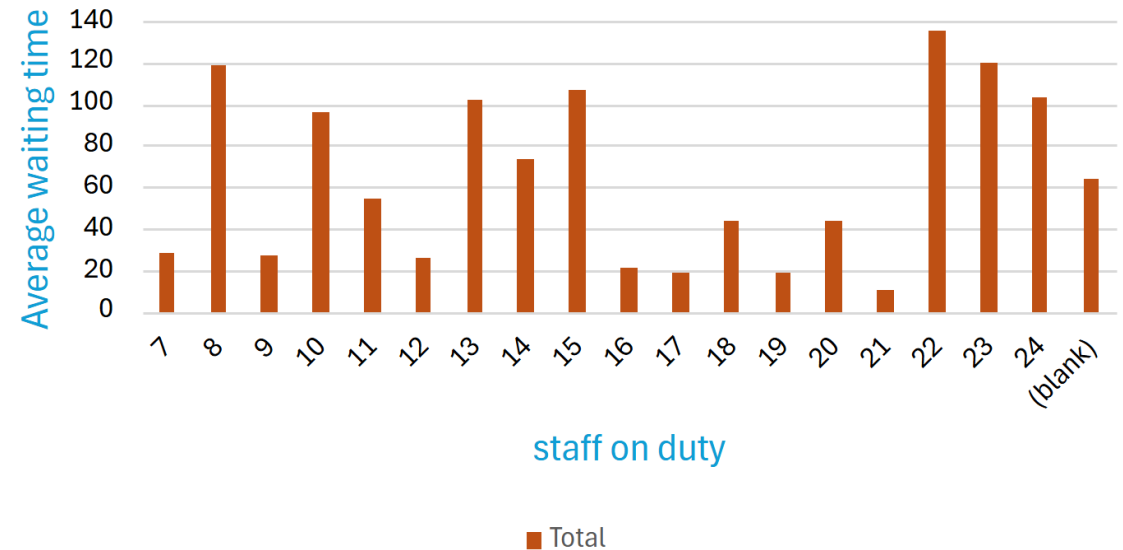


Staffing levels vs waiting time



- Higher staffing levels do not always lead to shorter waiting times.
- Lower staffing levels can be associated with shorter waits in some cases.
- Waiting times might be influenced by other factors (Patient arrival volumes, emergency demand, department workload, patient complexity, etc...)

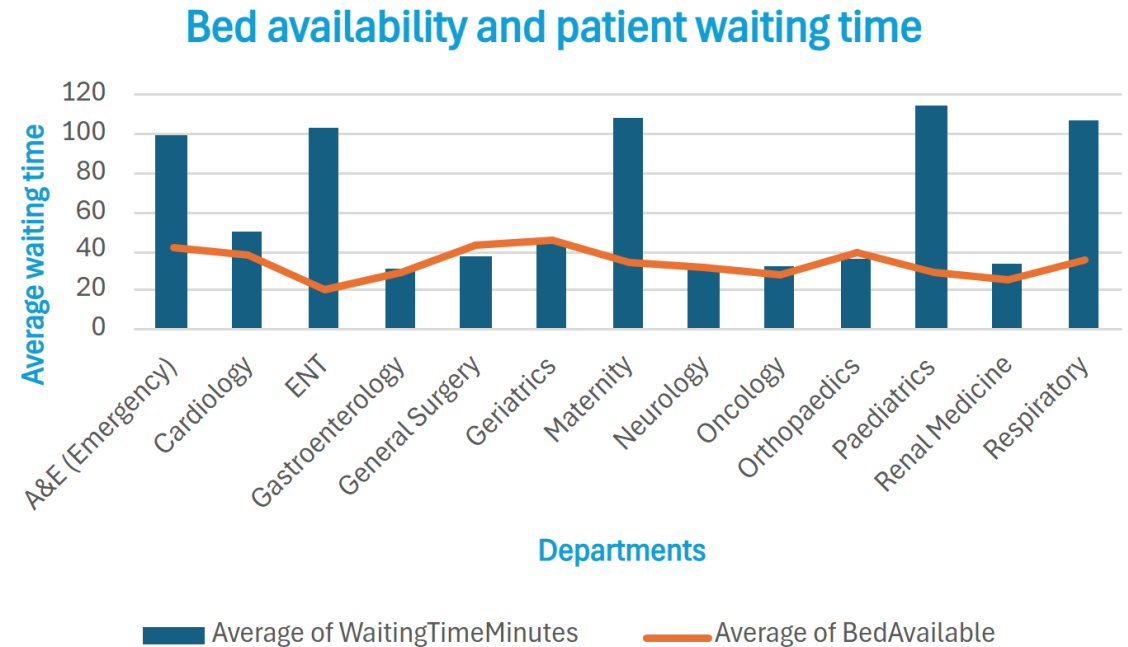
Average waiting time vs staffOnDuty



Bed Availability vs waiting time by department



- Some departments with more beds still experience long waiting times.
- Departments with lower waiting times generally have adequate bed capacity.
- Bed availability may influence delays, but it is not the only factor affecting patient flow.



Conclusion

- Waiting time problems are concentrated in specific departments, especially **Paediatrics**, **Maternity**, **Respiratory**, **ENT**, and **A&E**.
- Most patients were discharged successfully, but delays appear within the patient pathway.
- Staffing levels alone do not explain waiting time variation.
- The hospital should prioritise high-wait departments and ensure resources are allocated where they are most needed.



Recommendations

- Prioritize improvement in **departments with the longest waiting times** (Paediatrics, Maternity, Respiratory, ENT, A&E).
- Review **staffing capacity** and allocate clinical resources based on demand.
- Ensure the **right medical teams** are available when demand is high.



Recommendation



THANK YOU

FOR YOUR ATTENTION





Questions

Remarks



Suggestions